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Meter box ground clamp configured to prevent the clamp from moving off the meter box in response to a tightening torque

Abstract

A meter box ground clamp may include a body portion having a first portion, a second portion, and a gripping portion, and a fastener member configured to couple the body portion to an electrical meter box. The fastener member may be configured to extend through the first portion to engage a first surface of the meter box and move the first portion away from the first surface and move the gripping portion into engagement with a second surface of the meter box that faces away from the first surface. The receiving portion may be configured to receive the fastener member so as to create a force that urges the second portion of the body portion toward the meter box in response to a tightening torque applied to the fastener member so as to prevent the body portion from moving off the meter box in response to the tightening torque.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 63/295,601, filed Dec. 31, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to devices for implementing a ground connection between an electrical meter box and a common ground point and, more particularly, to clamps that mount to an electrical meter box and connect with a common ground point.

BACKGROUND

(2) Conventional devices for connecting a ground wire with an electrical meter typically include a clamp assembly that is clamped onto the exterior of the electrical meter box which is mounted on the wall of a building. Conventional clamp assemblies typically utilize a single hooking feature to engage the exterior of the meter box which does not provide the clamp assemblies with the flexibility to clamp to adapt to various mounting surfaces.

(3) When attaching conventional clamping assemblies to a meter box, technicians often use an impact driver, which can exert upwards of 150 in-lbs of torque on the screw of the ground clamp. Such torque can cause the entire clamp to rotate and walk off or jump off or move off the meter box.

(4) Conventional clamping assemblies are sometimes attached to a meter box that has been painted or otherwise includes a substance on its surface that prevents a clamping bolt from obtaining a strong ground connection with the meter box.

(5) It may be desirable to provide a meter box ground clamp that does not walk off the meter box while being tightened. It may be desirable to provide a meter box ground clamp having separate hooking features to allow flexibility when attaching the clamp to a surface. It may be desirable to provide a screw configured to cut through paint on a surface of the meter box to improve the ground connection.

SUMMARY

(6) According to various embodiments of the disclosure, a meter box ground clamp may include a clamp body, a clamp screw configured to couple the clamp body to an electrical meter box, and a ground engagement screw configured to couple a ground wire with the electrical meter box. The clamp body may comprise an L-shaped body that includes a first arm portion and a second arm portion that are configured as a monolithic structure of unitary construction, and the clamp screw may include a head portion and a threaded portion extending from the head portion. The first arm portion may include a first threaded opening configured to threadedly receive the threaded portion of the clamp screw, the head portion of the clamp screw may include an opening configured to extend through the head portion and to receive the ground wire therethrough, a top surface of the head portion may include a second threaded opening configured to extend into the opening and to threadedly receive the ground engagement screw, and the ground engagement screw may be configured to be threaded into the opening to engage the ground wire and mechanically and electrically couple the ground wire to the clamp body. The clamp body may include a gripping claw configured to extend from the second arm portion and curve toward the first arm portion, the gripping claw may be configured to engage a first surface of an electrical meter box, and the first arm portion may be configured to be positioned adjacent a second surface of the meter box. The clamp screw may be configured to be threaded into the first threaded opening to engage the second surface of the meter box that faces away from the first surface of the meter box so as to move the first arm portion away from the second surface of the meter box and move the gripping claw into engagement with the first surface of the meter box. The clamp screw may include a tip portion that extends from a radial center of the threaded portion at the free end and is configured to extend beyond the free end of the threaded portion such that the tip portion is configured to engage the second surface of the meter box before the threaded portion when the clamp body is being clamped to the meter box, and the tip portion may be configured to center the clamp screw at a desired location at the second surface of the meter box when the clamp screw is tightening the clamp body to the meter box. The threaded portion of the clamp screw may include a fluted portion at the free end that is configured to cut into the second surface of the meter box to strip paint from the second surface when the clamp screw is tightening the clamp body to the meter box so as to expose base metal of the meter box to facilitate a better grounding connection between the ground clamp and the meter box. The first threaded opening may be configured to receive the clamp screw at a first angle less than 90 degrees relative to a plane of the first arm portion such that a free end of the clamp screw forms a second angle greater than zero degrees relative to the second surface of the meter box, and the second angle may be configured to create a force that urges the second arm of the clamp body toward the meter box in response to a tightening torque applied to the clamp screw so as to prevent the clamp body from moving off the meter box in response to the tightening torque.

(7) According to various embodiments of the disclosure, a meter box ground clamp may include a

body portion including a first arm portion and a second arm portion, a fastener member configured to couple the body portion to an electrical meter box, and a ground portion configured to couple a ground wire with the meter box. The first arm portion may include an opening configured to receive the fastener member, and the body portion may include a gripping portion configured to extend from the second arm portion and curve toward the first arm portion. The fastener member may be configured to extend through the opening to engage a first surface of the meter box so as to move the first arm portion away from the first surface of the meter box and move the gripping portion into engagement with a second surface of the meter box that faces away from the first surface of the meter box, and the opening may be configured to receive the fastener member at a first angle less than 90 degrees relative to a plane of the first arm portion such that a free end of the fastener member forms a second angle greater than zero degrees relative to the second surface of the meter box. The second angle may be configured to create a force that urges the second arm portion of the body portion toward the meter box in response to a tightening torque applied to the fastener member so as to prevent the body portion from moving off the meter box in response to the tightening torque.

(8) According to various embodiments of the disclosure, a meter box ground clamp may include a body portion having a first portion, a second portion, and a gripping portion that is configured to extend from the second portion and a fastener member configured to couple the body portion to an electrical meter box. The fastener member may be configured to extend through a receiving portion of the first portion to engage a first surface of the meter box so as to move the first portion away from the first surface of the meter box and move the gripping portion into engagement with a second surface of the meter box that faces away from the first surface of the meter box, and the receiving portion may be configured to receive the fastener member at a first angle less than 90 degrees relative to a plane of the first portion such that a free end of the fastener member forms a second angle greater than zero degrees relative to the second surface of the meter box. The second angle may be configured to create a force that urges the second portion of the body portion toward the meter box in response to a tightening torque applied to the fastener member so as to prevent the body portion from moving off the meter box in response to the tightening torque.

(9) The foregoing and other features of construction and operation of the invention will be more readily understood and fully appreciated from the following detailed disclosure, taken in conjunction with accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front perspective view of an exemplary ground clamp, in accordance with various aspects of the disclosure, coupled with a meter box;
- (2) FIG. 2 is bottom perspective view of the exemplary ground clamp and meter box of FIG. 1;
- (3) FIG. 3 is an enlarged side view of the exemplary ground clamp of FIG. 1 and a portion of the meter box of FIG. 1;
- (4) FIG. 4 is side view of an exemplary clamp screw of the exemplary ground clamp FIG. 1;
- (5) FIG. 5 is a perspective view of the exemplary clamp screw of FIG. 4;
- (6) FIG. 6 is a perspective cross-sectional view of the exemplary ground clamp of FIG. 1 and a portion of the meter box of FIG. 1;
- (7) FIG. 7 is a perspective view of another exemplary ground clamp, in accordance with various aspects of the disclosure;
- (8) FIG. 8 is a front view of the exemplary ground clamp of FIG. 7;
- (9) FIG. 9 is a side view of the exemplary ground clamp of FIG. 7; and
- (10) FIG. 10 is a perspective view of another exemplary ground clamp, in accordance with various

aspects of the disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

(11) As a preface to the detailed description, it should be noted that, as used in this specification and the appended claims, the singular forms “a”, “an” and “the” include plural referents, unless the context clearly dictates otherwise.

(12) FIGS. **1-3** illustrate an exemplary meter box ground clamp **100** in accordance with various aspects of the enclosure. The ground clamp **100** includes a clamp body **120**, a clamp screw **140**, and a ground engagement screw **160** that facilitate connecting a ground wire **170** to an enclosure of an electrical meter box. The ground clamp **100** may be configured to mount the ground wire **170** to a conventional, rectangular electrical meter box **180** having a drip cap **182**, a front panel **184**, a rear panel (not shown), and side panels **186** (only one shown). The drip cap **182** may include a vertically extending lip portion **190** that overhangs and overlaps the top portions of the panels **184**, **186**.

(13) As illustrated in FIG. **3**, the clamp body **120** is substantially L-shaped with a first arm portion **122** and a second arm portion **124**. The first arm portion **122** and the second arm portion **124** may be a monolithic structure of unitary construction, with the arm portions **122**, **124** being bent into the L-shaped configuration. The first arm portion **122** includes a threaded opening **123** configured to threadedly receive the clamp screw **140**, which is configured to tighten the clamp body **120** to the meter box **180**.

(14) The clamp screw **140** includes a head portion **142** and a threaded portion **144** extending from the head portion **142**. The head portion **142** may be hexagonal or square to facilitate tightening by a wrench or other torque applying tool. The head portion **142** includes an opening **146** extending through the opposite side walls **148** of the head portion **142**. For example, the opening **146** may extend through the head portion **142** via opposite walls of a hexagonal shaped head portion. The opening **146** is sized to receive the ground wire **170** therethrough, shown in FIG. **1**.

(15) A top surface **148** of the head portion **142** includes a threaded opening **150** that extends into the opening **146**. The ground engagement screw **160** is configured to be threadedly coupled with the opening **150** such that the ground engagement screw **160** can be adjusted to extend into the opening **146** to engage the ground wire **170** extending through the opening **146** and force the ground wire **170** against a wall **147** of the opening to mechanically and electrically couple the ground wire **170** to the clamp body **120**. The ground engagement screw **160** includes a head portion **162** and a threaded portion **164** configured to be threadedly coupled with the opening **150**. The head portion **162** includes one or more structural features configured to receive a tightening tool, such as a screwdriver, a wrench, or the like.

(16) The clamp body **120** includes a pair of gripping claws **130** that extend from the second arm portion **124**. The gripping claws **130** and the second arm portion **124** may be a monolithic structure of unitary construction, with the gripping claws **130** being cut from the second arm portion **124** and formed into a curved shape to define the gripping claws **130**, as shown in FIG. **2**. For example, three sides of each gripping claw **130** are cut from the second arm portion **124** such that the gripping claws **130** remain connected to the second arm portion **124** in a cantilever manner. The gripping claws **130** are configured to curve from the second arm portion **124** toward the first arm portion **122**. Ends **131** of the gripping claws **130** are configured to engage a surface **191** of the lip portion **190** of the drip cap **182**.

(17) In use, the first arm portion **122** is positioned adjacent a top surface **183** of the drip cap **182**, and the clamp screw **140** is withdrawn from the threaded opening **150** by an amount sufficient to permit the gripping claws **130** to be positioned at an opposite side of the surface **191** of the lip portion **190** relative to the first arm portion **122**. The clamp screw **140** is threaded into the opening **150** to engage the top surface **183** of the drip cap **182**. Further threading of the clamp screw **140** into the opening **150** moves the first arm portion **122** away from the top surface **183** of the drip cap **182** and moves the gripping claws **130** toward and into engagement with the surface **191** of the lip

portion **190**. The clamp screw **140** can be threaded into the opening **150** until the clamp body **120** is secured to the drip cap **182** with a desired gripping force. The grounding wire **170** is inserted through the head portion **142** of the clamp screw **140** via the opening **146**, and the ground engagement screw **160** is threaded into the opening **150** to mechanically and electrically couple the ground wire **170** to the clamp body **120**. The separate gripping claws **130** have multiple contact regions **132** configured to engage the surface **191** of the lip portion **190**. In some aspects, the contact regions **132** may be angled (i.e., not parallel) relative to one another such that the gripping claws **130** are configured to engage the surface **191** of the lip portion **190** even if the surface **190** is nonlinear. Consequently, the gripping claws **130** are configured to adapt to an inconsistent (e.g., non-linear) surface of the lip portion **190** to maintain a secure connection between the ground clamp **100** and the meter box **180**.

(18) In some embodiments of the ground clamp **100**, the clamp screw **140** may be configured as the clamp screw **440** shown in FIGS. 4-6. As shown in FIGS. 4-6, clamp screw **440** is configured similar to a self-tapping sheet metal screw. For example, the clamp screw **440** includes a head portion **442** and a threaded portion **444** extending from the head portion **442** to a free end **445**. The head portion **442** may be hexagonal or square to facilitate tightening by a wrench or other torque applying tool. The head portion **442** includes an opening **446** extending through the opposite side walls **448** of the head portion **442**. For example, the opening **446** may extend through the head portion **442** via opposite walls **448** of a hexagonal shaped head portion. The opening **446** is sized to receive the ground wire **170** therethrough, as described above.

(19) As illustrated in FIGS. 4-6, the clamp screw **440** includes a tip portion **452** that extends from a radial center of the threaded portion **444** at the free end **445**. As best shown in FIGS. 4 and 6, the tip portion **452** extends beyond the free end **445** of the threaded portion **444**. The clamp screw **440** may also include flutes **454** cut into the threaded portion **444** at the free end **445**. The flutes **454** may be configured as wedges cut into the threaded portion **444** at about a 45 degree angle.

(20) The tip portion **452** is configured to engage the top surface **183** of the drip cap **182** before the threaded portion when the clamp body **120** is being clamped to the drip cap **182**. As such, the tip portion **452** is configured to center the clamp screw **440** at a desired location at the top surface **183** of the drip cap **182** as torque is applied to the head portion **442** and to prevent the ground clamp **100** from walking (i.e., moving) relative to the drip cap **182** while the clamp body is being tightened to the drip cap **182**. The flutes **445** are configured to cut into the top surface **183** of the drip cap **182** to strip any paint or other substance off the drip cap **182**. For example, the wedge-shaped flutes create a sharp/blunt edge configured to scrape paint or other substance off the top surface **183** to expose the base metal of the drip cap **182** to facilitate better a better grounding connection between the ground clamp **100** and the meter box **180**.

(21) Referring now to FIGS. 7-9, another embodiment of an exemplary ground clamp **700** is illustrated. The ground clamp **700** is similar to the ground clamp **100** in that the ground clamp includes a clamp body **720**, a clamp screw **740**, and a ground engagement screw **760** which facilitate connecting a ground wire **170** to an enclosure of an electrical meter box. The ground clamp **700** may be configured to mount the ground wire **170** to the conventional, rectangular electrical meter box described above.

(22) As illustrated in FIGS. 7 and 9, the clamp body **720** is substantially L-shaped with a first arm portion **722** and a second arm portion **724**. The first arm portion **722** and the second arm portion **724** may be a monolithic structure of unitary construction, with the arm portions **722**, **724** being bent into the L-shaped configuration. The first arm portion **722** includes a threaded opening **723** configured to threadedly receive the clamp screw **740**, which is configured to tighten the clamp body **720** to the meter box **180**.

(23) The clamp screw **740** includes a head portion **742** and a threaded portion **744** extending from the head portion **742**. The head portion **742** may be hexagonal or square to facilitate tightening by a wrench or other torque applying tool. The head portion **742** includes an opening **746** extending

through the opposite side walls **748** of the head portion **742**. For example, the opening **746** may extend through the head portion **742** via opposite walls of a hexagonal shaped head portion. The opening **746** is sized to receive the ground wire **170** therethrough, as discussed above.

(24) A top surface **748** of the head portion **742** includes a threaded opening **750** that extends into the opening **746**. The ground engagement screw **760** is configured to be threadedly coupled with the opening **750** such that the ground engagement screw **760** can be adjusted to extend into the opening **746** to engage the ground wire **170** extending through the opening **746** and force the ground wire **170** against a wall **747** of the opening to mechanically and electrically couple the ground wire **170** to the clamp body **720**. The ground engagement screw **760** includes a head portion **762** and a threaded portion **764** configured to be threadedly coupled with the opening **750**. The head portion **762** includes one or more structural features configured to receive a tightening tool, such as a screwdriver, a wrench, or the like.

(25) The clamp body **720** includes a pair of gripping claws **730** that extend from the second arm portion **724**. The gripping claws **730** and the second arm portion **724** may be a monolithic structure of unitary construction, with the gripping claws **730** being cut from the second arm portion **724** and formed into a curved shape to define the gripping claws **730**, as shown in FIG. 7. For example, three sides of each gripping claw **730** are cut from the second arm portion **724** such that the gripping claws **730** remain connected to the second arm portion **724** in a cantilever manner. The gripping claws **730** are configured to curve from the second arm portion **724** toward the first arm portion **722**. Ends **731** of the gripping claws **730** are configured to engage a surface **191** of the lip portion **190** of the drip cap **182**.

(26) As best illustrated in FIG. 8, the threads of the threaded opening **723** are configured to receive the clamp screw **740** at an angle α relative to a plane of the first arm portion **722**. In various aspects, the angle α may range from 45 degrees to 85 degrees. Consequently, the free end **745** of the clamp screw **740** forms an angle β relative to the plane. In various aspects, the angle β may range from 5 degrees to 45 degrees. As a result of the angle β between the free end **745** of the clamp screw **740** and the plane, when the clamp screw **740** is threaded into the threaded opening **750** to tighten the clamp body **720** to the drip cap **182**, the tightening torque creates a force that urges the clamp body **720** in the direction F in FIG. 9 (out of the paper in FIG. 8), which is toward the meter box **180**. Thus, if the clamp body **720** tends to walk or travel relative to the meter box **180**, the tightening torque will urge the clamp body **720** toward the meter box **180** rather than off the meter box **180**.

(27) However, depending on the angle β , if the clamp screw **740** is configured similar to the clamp screw **440** described above, the tip portion **752** may or may not be configured to engage the top surface **183** of the drip cap **182** before the threaded portion **744** when the clamp body **120** is being clamped to the drip cap **182**. For example, if the angle β is greater than a predetermined angle, the tip portion **752** will not be configured to engage the top surface **183** of the drip cap **182** before the threaded portion **744** when the clamp body **120** is being clamped to the drip cap **182**. But if the angle β is less than or equal to the predetermined angle, the tip portion **752** will be configured to engage the top surface **183** of the drip cap **182** before the threaded portion **744** when the clamp body **120** is being clamped to the drip cap **182**.

(28) In use, the first arm portion **722** is positioned adjacent the top surface **183** of the drip cap **182**, and the clamp screw **740** is withdrawn from the threaded opening **750** by an amount sufficient to permit the gripping claws **730** to be positioned at an opposite side of the surface **191** of the lip portion **190** relative to the first arm portion **722**. The clamp screw **740** is threaded into the opening **750** at the angle β to engage the top surface **183** of the drip cap **182**. Further threading of the clamp screw **740** into the opening **750** moves the first arm portion **722** away from the top surface **183** of the drip cap **182** and moves the gripping claws **730** toward and into engagement with the surface **191** of the lip portion **190**. The clamp screw **740** can be threaded into the opening **750** until the clamp body **720** is secured to the drip cap **182** with a desired gripping force. As discussed above,

the tightening torque may create a force that urges the clamp body **720** in the direction F in FIG. **9** (out of the paper in FIG. **8**), which is toward the meter box **180**. Thus, if the clamp body **720** tends to walk or travel relative to the meter box **180**, the tightening torque will urge the clamp body **720** toward the meter box **180** and prevent the clamp body **720** from walking off the meter box **180**.

(29) The grounding wire **170** is inserted through the head portion **742** of the clamp screw **740** via the opening **746**, and the ground engagement screw **760** is threaded into the opening **750** to mechanically and electrically couple the ground wire **170** to the clamp body **720**. The separate gripping claws **730** have multiple contact regions **732** configured to engage the surface **191** of the lip portion **190**. In some aspects, the contact regions **732** may be angled (i.e., not parallel) relative to one another such that the gripping claws **730** are configured to engage the surface **191** of the lip portion **190** even if the surface **191** is nonlinear. Consequently, the gripping claws **730** are configured to adapt to an inconsistent (e.g., non-linear) surface of the lip portion **190** to maintain a secure connection between the ground clamp **700** and the meter box **180**.

(30) While this invention has been described in terms of several preferred embodiments, there are alteration, permutations, and equivalents, which fall within the scope of this invention. It should also be noted that there are many alternative ways of implementing the methods and apparatuses of the present invention. It is therefore intended that the following appended claims be interpreted as including all such alterations, permutations, and equivalents as fall within the true spirit and scope of the present invention.

Claims

1. A meter box ground clamp configured to prevent the clamp from moving off the meter box in response to a tightening torque, comprising: a clamp body; a clamp screw configured to couple the clamp body to an electrical meter box; a ground engagement screw configured to couple a ground wire with the electrical meter box; wherein the clamp body comprises an L-shaped body that includes a first arm portion and a second arm portion that are configured as a monolithic structure of unitary construction; wherein the clamp screw includes a head portion and a threaded portion extending from the head portion; wherein the first arm portion includes a first threaded opening configured to threadedly receive the threaded portion of the clamp screw; wherein the head portion of the clamp screw includes an opening configured to extend through the head portion and to receive the ground wire therethrough; wherein a top surface of the head portion includes a second threaded opening configured to extend into the opening and to threadedly receive the ground engagement screw; wherein the ground engagement screw is configured to be threaded into the opening to engage the ground wire and mechanically and electrically couple the ground wire to the clamp body; wherein the clamp body includes a gripping claw configured to extend from the second arm portion and curve toward the first arm portion; wherein the gripping claw is configured to engage a first surface of an electrical meter box, and the first arm portion is configured to be positioned adjacent a second surface of the meter box; wherein the clamp screw is configured to be threaded into the first threaded opening to engage the second surface of the meter box that faces away from the first surface of the meter box so as to move the first arm portion away from the second surface of the meter box and move the gripping claw into engagement with the first surface of the meter box; wherein the clamp screw includes a tip portion that extends from a radial center of the threaded portion at the free end and is configured to extend beyond the free end of the threaded portion such that the tip portion is configured to engage the second surface of the meter box before the threaded portion when the clamp body is being clamped to the meter box; wherein the tip portion is configured to center the clamp screw at a desired location at the second surface of the meter box when the clamp screw is tightening the clamp body to the meter box; wherein the threaded portion of the clamp screw includes a fluted portion at the free end that is configured to cut into the second surface of the meter box to strip paint from the second surface when the clamp

screw is tightening the clamp body to the meter box so as to expose base metal of the meter box to facilitate a better grounding connection between the ground clamp and the meter box; wherein the first threaded opening is configured to receive the clamp screw at a first angle less than 90 degrees relative to a plane of the first arm portion such that a free end of the clamp screw forms a second angle greater than zero degrees relative to the second surface of the meter box; and wherein the second angle is configured to create a force that urges the second arm of the clamp body toward the meter box in response to a tightening torque applied to the clamp screw so as to prevent the clamp body from moving off the meter box in response to the tightening torque.

2. The clamp of claim 1, wherein the gripping claw comprises two gripping claws spaced apart from one another.

3. The clamp of claim 1, wherein each of the gripping claws has a contact region configured to engage the first surface of the meter box to maintain a secure connection between the ground clamp and the meter box.

4. A meter box ground clamp configured to prevent the clamp from moving off the meter box in response to a tightening torque, comprising: a body portion including a first arm portion and a second arm portion; a fastener member configured to couple the body portion to an electrical meter box; a ground portion configured to couple a ground wire with the meter box; wherein the first arm portion includes an opening configured to receive the fastener member; wherein the body portion includes a gripping portion configured to extend from the second arm portion and curve toward the first arm portion; wherein the fastener member is configured to extend through the opening to engage a first surface of the meter box so as to move the first arm portion away from the first surface of the meter box and move the gripping portion into engagement with a second surface of the meter box that faces away from the first surface of the meter box; wherein the opening is configured to receive the fastener member at a first angle less than 90 degrees relative to a plane of the first arm portion such that a free end of the fastener member forms a second angle greater than zero degrees relative to the second surface of the meter box; and wherein the second angle is configured to create a force that urges the second arm portion of the body portion toward the meter box in response to a tightening torque applied to the fastener member so as to prevent the body portion from moving off the meter box in response to the tightening torque.

5. The clamp of claim 4, wherein the grounding portion comprises a ground engagement screw configured to couple a ground wire with the electrical meter box.

6. The clamp of claim 5, wherein the fastener member comprises a clamp screw that includes a head portion and a threaded portion extending from the head portion.

7. The clamp of claim 6, wherein the head portion of the clamp screw includes an opening configured to extend through the head portion and to receive a ground wire therethrough, a top surface of the head portion includes a threaded opening configured to extend into the opening and to threadedly receive the ground engagement screw, and the ground engagement screw is configured to be threaded into the opening in the head portion to engage the ground wire and mechanically and electrically couple the ground wire to the body portion.

8. The clamp of claim 4, wherein the opening in the first arm portion comprises a threaded opening that is configured to threadedly receive a threaded portion of the fastener member.

9. The clamp of claim 4, wherein the body portion comprises an L-shaped body that includes the first arm portion and the second arm portion configured as a monolithic structure of unitary construction.

10. The clamp of claim 4, wherein the fastener member comprises a screw that includes a threaded portion and a tip portion configured to extend from a radial center of the threaded portion at a free end of the fastener member, and the tip portion is configured to extend beyond the free end of the threaded portion such that the tip portion is configured to engage the second surface of the meter box before the threaded portion when the body portion is being clamped to the meter box so as to center the screw at a desired location at the second surface of the meter box when the screw is

tightening the body portion to the meter box.

11. The clamp of claim 4, wherein the fastener member comprises a screw that includes a threaded portion and a fluted portion at a free end of the threaded portion, and the fluted portion is configured to cut into the second surface of the meter box to strip paint from the second surface when the clamp screw is tightening the body portion to the meter box so as to expose a base metal of the meter box to facilitate an improved grounding connection between the ground clamp and the meter box.

12. A meter box ground clamp configured to prevent the clamp from moving off the meter box in response to a tightening torque, comprising: a body portion including a first portion, a second portion, and a gripping portion that is configured to extend from the second portion; a fastener member configured to couple the body portion to an electrical meter box; wherein the fastener member is configured to extend through a receiving portion of the first portion to engage a first surface of the meter box so as to move the first portion away from the first surface of the meter box and move the gripping portion into engagement with a second surface of the meter box that faces away from the first surface of the meter box; wherein the receiving portion is configured to receive the fastener member at a first angle less than 90 degrees relative to a plane of the first portion such that a free end of the fastener member forms a second angle greater than zero degrees relative to the second surface of the meter box; and wherein the second angle is configured to create a force that urges the second portion of the body portion toward the meter box in response to a tightening torque applied to the fastener member so as to prevent the body portion from moving off the meter box in response to the tightening torque.

13. The clamp of claim 12, further comprising a ground portion that is configured to couple a ground wire with the meter box.

14. The clamp of claim 13, wherein the grounding portion includes a ground engagement screw configured to couple a ground wire with the electrical meter box.

15. The clamp of claim 14, wherein the fastener member comprises a clamp screw that includes a head portion and a threaded portion extending from the head portion.

16. The clamp of claim 15, wherein the head portion of the clamp screw includes a receiving portion configured to extend through the head portion and to receive a ground wire therethrough, a top surface of the head portion includes an opening configured to extend into the receiving portion and to threadedly receive the ground engagement screw, and the ground engagement screw is configured to be threadedly received by the opening in the head portion to engage the ground wire and mechanically and electrically couple the ground wire to the body portion.

17. The clamp of claim 12, wherein the opening in the first arm portion comprises a threaded opening configured to threadedly receive a threaded portion of the fastener member.

18. The clamp of claim 12, wherein the body portion comprises an L-shaped body that includes the first portion and the second portion configured as a monolithic structure of unitary construction.

19. The clamp of claim 12, wherein the fastener member comprises a screw that includes a threaded portion and a tip portion configured to extend from a radial center of the threaded portion at a free end of the fastener member, and the tip portion is configured to extend beyond the free end of the threaded portion such that the tip portion is configured to engage the second surface of the meter box before the threaded portion when the body portion is being clamped to the meter box so as to center the screw at a desired location at the second surface of the meter box when the screw is tightening the body portion to the meter box.

20. The clamp of claim 12, wherein the fastener member comprises a screw that includes a threaded portion and a fluted portion at a free end of the threaded portion, and the fluted portion is configured to cut into the second surface of the meter box to strip paint from the second surface when the clamp screw is tightening the body portion to the meter box so as to expose a base metal of the meter box to facilitate an improved grounding connection between the ground clamp and the meter box.

